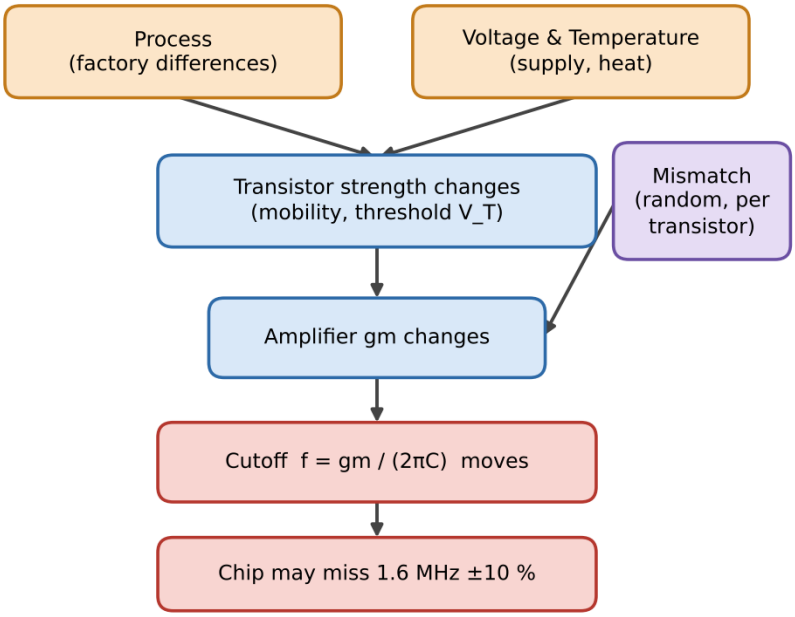
The Tale of the Fussy Filter

How we asked a simple question — *“how many chips actually work?”* — and found out why



A plain-language, story-style guide to the project

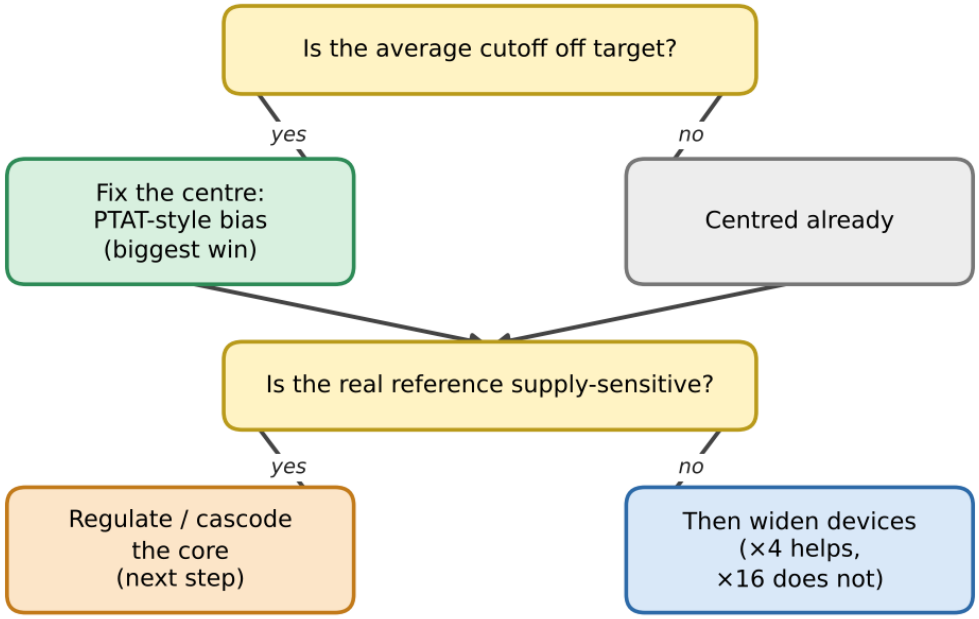
Statistical Characterization of a Gm-C Continuous-Time Filter in SKY130: Monte Carlo and PVT-Corner Study

Jaidev K · S Jai Akaash · Mohanakrishnan S · Jeswant P M
VIT Chennai · Guide: Dr. Brintha Therese A · BAECE203 Analog Electronics

HOW TO READ THIS

The story in one minute

Read it like a story. Every chapter has the same pattern: a **scene** (what is going on, in everyday words), a **flowchart** (the logic in boxes), and a **“what we actually did”** box (the real engineering). New words are explained the first time they appear, and there is a glossary at the end.



The whole project’s logic in one picture — keep coming back to this one.

Chapter	The scene	The real thing
1	A radio that will not stay tuned	Cutoff frequency drifts with process, voltage, temperature
2	The cast: a tap, a bucket, a helper	The 3-OTA Gm-C biquad and its OTA cell
3	The Heat villain and the clever pedal (PTAT)	Temperature compensation by bias current
4	The Wobble twin and the coin rule (Pelgrom)	Random mismatch and why area helps
5	Three ways to feed the helpers	Fixed voltage, ideal PTAT, real beta-multiplier core
6	The great rehearsal	45 conditions × 750 Monte Carlo trials
7	The results	Yield 22.4 % / 88.7 % / 71.7 %
8	Why the real core lost 17 points	Supply sensitivity
9	Did we cheat ourselves?	Five cross-checks
10	What we learned, and what is next	Guidelines, limits, deliverables

CHAPTER 1

The Radio That Would Not Stay Tuned

Why this project exists at all

Imagine you tune a guitar string to the perfect note. Then someone warms the room, and the note slides sharp. You tune it again — but a second guitar, *made on the same day by the same maker*, is still a bit different. Electronic filters have exactly this problem.

A **filter** is a circuit that lets some signals through and blocks others. Ours is a **low-pass** filter: it lets everything *below* 1.6 MHz pass and blocks what is above. The point where it starts to block is the **cutoff frequency**. Think of it as the note the guitar string is tuned to.

Our filter is built from **amplifiers** and **capacitors** (no resistors). The cutoff is simply **f = gm / (2π·C)**. The amplifier's strength **gm** sits in the numerator — so if the amplifier gets weaker or stronger, the cutoff slides. And the amplifier's strength is not stable. It changes with four things:

Villain	What it is	Everyday picture
Process	Factory-to-factory and batch-to-batch differences	Guitars from different batches
Voltage	The power-supply level wobbles ±10 %	Battery running low
Temperature	The chip gets cold or hot (−20 to 85 °C)	Warm room
Mismatch	Even "identical" transistors on one chip differ randomly	Two strings from the same pack

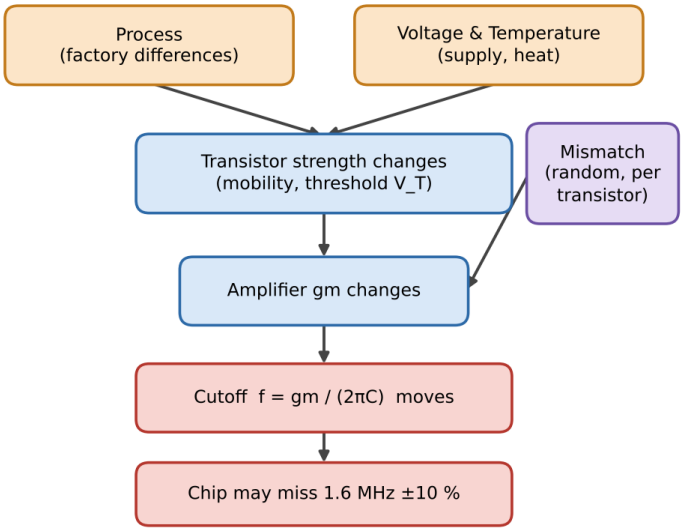


Figure 1 — The chain of cause and effect. Everything on top ends up as a wrong cutoff at the bottom.

The question we asked
Suppose we make thousands of these filters and they must meet the spec: **cutoff = 1.6 MHz ±10 %** *and* **sharpness Q between 0.6 and 0.8**. **What fraction actually passes — and how much does the way we feed current to the amplifiers change that?** That fraction is called **yield**.

Why it is new
Studies like this existed only for private, paid factory processes. We did it in **SKY130**, a **free, open-source** process — and did it at the *whole filter* level, not just a single amplifier. That gap is our contribution.

CHAPTER 2

Meet the Cast

A tap, a bucket, and a second bucket

Picture a garden tap that pours into a bucket. The harder you turn the tap, the faster the bucket fills. A **transconductance amplifier (OTA)** is that tap: you give it a *voltage* and it pours out a proportional *current*. The **capacitor** is the bucket: it collects that current as stored charge. Fill a bucket slowly and the level rises slowly — that is what slows down fast signals, i.e. a low-pass filter.

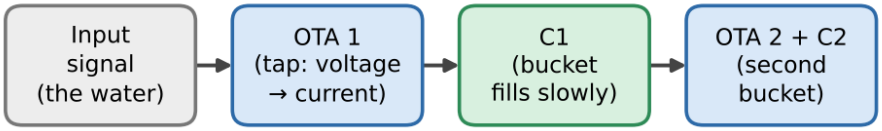


Figure 2 — The water analogy: input → tap → bucket → tap → bucket.

Our filter uses **three** taps and **two** buckets:

Part	Job	Plain meaning
OTA 1 + C1 (10 pF)	First integrator	Tap 1 filling bucket 1
OTA 2 + C2 (10 pF)	Second integrator	Tap 2 filling bucket 2 — gives a steeper roll-off (“second-order”)
OTA 3	Acts as a resistor $1/gm_3$	A small leak that stops the filter from ringing — it sets **Q**
Feedback wire	Output back to input	Makes the DC gain exactly 1

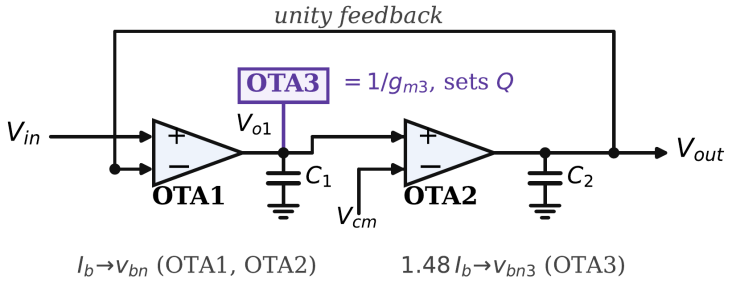


Figure 3 — The real circuit: the “biquad”. OTA1 and OTA2 are integrators; OTA3 is the damping resistor.

> **Two formulas to remember**
> **Cutoff:** $\omega_0 = gm / C$ (so f ≈ 1.6 MHz needs gm ≈ 100 µS with C = 10 pF)
> **Sharpness:** $Q = gm / gm_3$. We bias OTA3 at 1.48× the current of the others, giving Q ≈ 0.72 — just above the “flattest possible” Butterworth value of 0.707.

What is inside one OTA? Five transistors:

Transistors	Size (W/L µm)	Job
M1, M2 (NMOS pair)	4 / 0.5	Input pair — compares the two inputs

Transistors	Size (W/L µm)	Job
M3, M4 (PMOS mirror)	8 / 0.5	Load — copies current, sets the output’s resting level
M5 (NMOS tail)	6 / 1	**The tuning knob:** its gate voltage sets the current and therefore gm

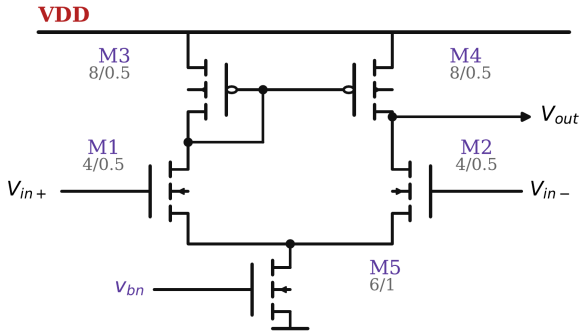


Figure 4 — One OTA cell. Only the tail bias differs between OTA1, OTA2 and OTA3.

What we actually did (Chapter 2)
Designed the 3-OTA biquad in SKY130A, chose sizes, then tuned them in simulation until the typical cutoff hit 1.6 MHz (measured 1.610 MHz, Q = 0.722). There is no common-mode feedback and no output buffer — a simplification we list as a limit.

CHAPTER 3

The Heat Villain and the Clever Pedal

PTAT — the first big idea

The first villain is **Heat**. Inside silicon, current is carried by electrons. When the chip is hot, the atoms jiggle more and electrons bump into them — like running through thicker mud. The technical word is that **mobility** falls. With the same gate voltage, a hot transistor passes *less* current, so the amplifier is *weaker*, so gm falls, and the cutoff slides down.

We saw this directly: with a fixed voltage bias at 1.8 V, the cutoff climbed from **1.20 MHz at −20 °C** to **1.94 MHz at 85 °C** (target 1.6). Across every condition the mean cutoff roamed from **0.67 to 2.54 MHz**.

The fix is a pedal. A car loses power going uphill, so you press the accelerator harder as the hill steepens and the speed stays constant. A **PTAT** current does the same: **P**roportional **T**o **A**bsolute **T**emperature — it *rises when it is hot*. The extra current exactly buys back the strength the heat took away.

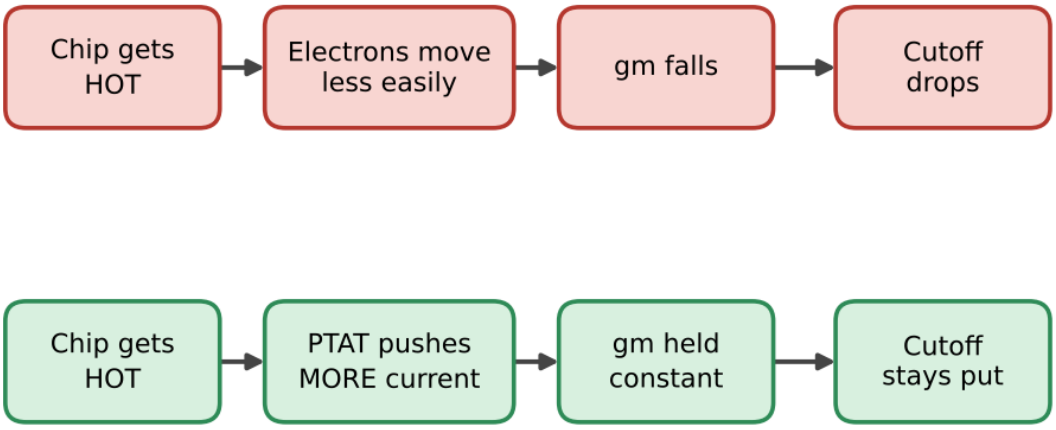


Figure 5 — Top row: no compensation. Bottom row: with PTAT bias the two effects cancel.

Question	Answer
Why not a plain PTAT (exponent 1)?	It overcorrects: the cutoff drifts about ∓6 % at −20 °C and 85 °C.
What did we use?	$I(T) = 12\ \mu A \times (T / 300.15\ K)^{0.6}$ — we scanned the exponent; 0.6 holds f_3 within ±1.5 % at all five corners.
Where does it come from physically?	A beta-multiplier circuit's current depends on a resistor and on mobility, so it naturally follows temperature.

What we actually did (Chapter 3)
Built two versions: an **ideal** PTAT (a perfect mathematical current source — the best-case bound) and a **transistor-level** beta-multiplier. We probed the real core from −40 to 125 °C at three corners and three supplies and *measured* its exponent: **0.64–0.67**, remarkably close to the 0.6 we tuned for.

CHAPTER 4

The Wobble Twin and the Coin Rule

Mismatch and Pelgrom — the second big idea

The second villain, **Mismatch**, is more mischievous because it is random. You draw two transistors identically, but the factory can only place doping atoms and cut edges approximately. Each transistor ends up with a slightly different **threshold voltage V_T** (the gate voltage at which it switches on). No setting of the bias can undo this, because every chip gets a *different* random draw.

Pelgrom's rule says how bad it is: **$\sigma(\Delta V_T) = A / \sqrt{W \cdot L}$**. In words: the random error shrinks as the transistor's area grows. Why? Flip one coin and you can easily get 100 % heads. Flip ten thousand coins and you will be extremely close to 50 %. A big transistor is made of many "coins", so its random errors average out.

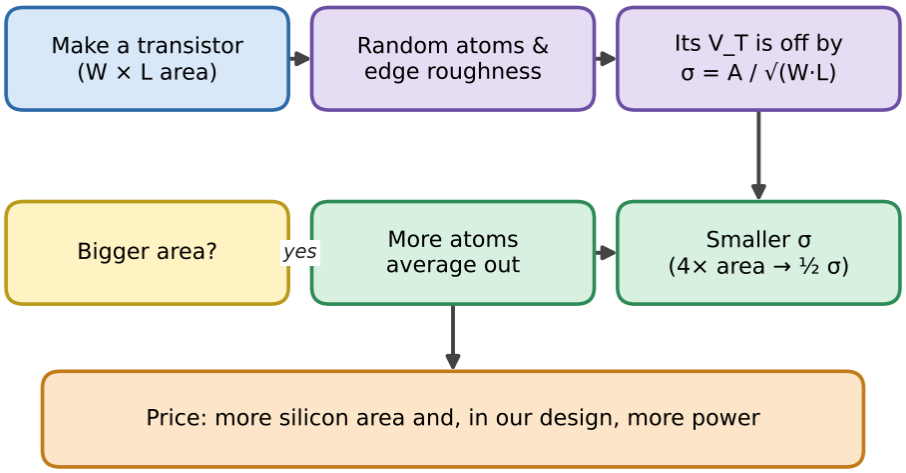


Figure 6 — Pelgrom in boxes. Area buys accuracy, but you pay in silicon and power.

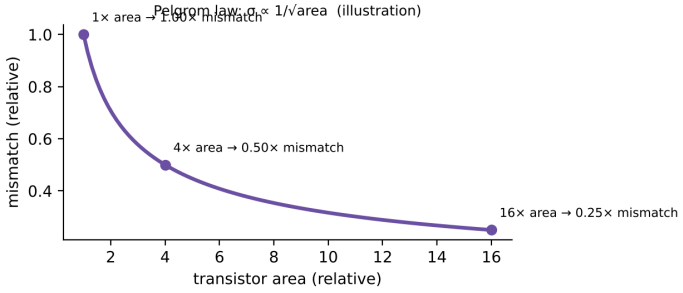


*Figure 7 — The square-root law: to halve the mismatch you need **four times** the area.*

Using Pelgrom as a lie detector. We ran 33,750 random simulated chips per design. How do we know those simulations are right? We recomputed the spread *by hand*, a completely different way:

Step	What we did	Result
1	Take each device group's size and use Pelgrom to get its random voltage error	2.76 mV input pair · 1.45 mV tail and reference · 2.89 mV load

Step	What we did	Result
2	Ask the simulator: “if this one device shifts 1 mV, how much does f_3 move?” (sensitivity)	Per-device sensitivities
3	Combine them (root-sum-square)	Predicted $\sigma(f_3)$ = **5.62 %**
4	Compare to the full Monte Carlo	Simulated $\sigma(f_3)$ = **5.91 %** ✔
5	Repeat for the ×4 bigger mirror	Predicted 4.87 % vs simulated 4.81 % ✔

The important twist
Mismatch makes about a **5–7 % random spread in cutoff**, and **no bias scheme removes it** — PTAT even leaves it very slightly larger (6.07 % vs 5.32 %). A reference can move where the cutoff *sits*; only transistor area can shrink how much it *wobbles*. So: **centre first, narrow second.**

CHAPTER 5

Three Ways to Feed the Helpers

The experiment's three contestants

Every OTA needs a current. How we supply it is the experiment. We built three versions of the filter — identical in every way except how the tail current is generated.

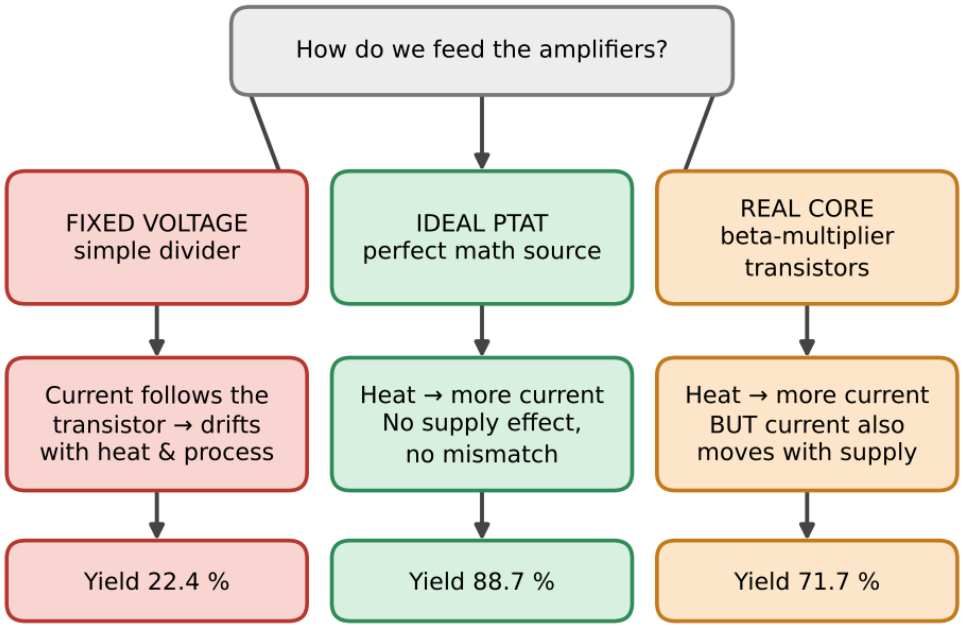


Figure 8 — Three designs, three outcomes (spoiler: see the last row).

Design	How it works	Strength	Weakness
1. Fixed voltage	A DC voltage straight onto each tail gate (0.70 V for OTA1/2, 0.73 V for OTA3)	Dead simple — what you get from a resistor divider	Current follows transistor behaviour, so every shift passes into gm
2. Ideal PTAT	A behavioural source, I = 12 µA·(T/300.15)$^{0.6}$, mirrored 1 : 1.48 for OTA3	Perfect temperature law, no supply effect, no mismatch	Not buildable — it is the upper bound
3. Real core	Beta-multiplier: PMOS mirror → two NMOS branches, one with a 51.5 kΩ poly resistor, K = 4, plus 5 MΩ start-up	Real transistors with their own mismatch and supply behaviour — honest	Current moves ≈ 0.5 % per 1 % of supply

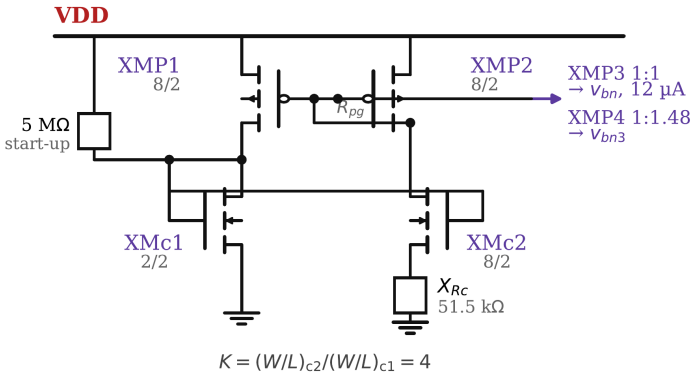


Figure 9 — The real transistor-level beta-multiplier core. Measured: exponent 0.64–0.67; current 11.3 / 12.0 / 12.3 µA at 1.62 / 1.8 / 1.98 V.

Why bother with the real core? If we had only the ideal source, a reviewer could say: “of course it looks good, your reference is perfect.” Putting a real, imperfect reference *inside* the simulation shows what you would truly get.

> **What we actually did (Chapter 5)**
> Designed the beta-multiplier (PMOS 8/2, NMOS 2/2 and 8/2, resistor length calibrated so it delivers 12.0 µA at tt, 1.8 V, 27 °C), two PMOS mirrors at 1:1 and 1:1.48 to feed OTA1/2 and OTA3, and a start-up path. Lengthening the core devices did not help, so L = 2 µm was kept.

CHAPTER 6

The Great Rehearsal

How we “made” 33,750 chips without a factory

We cannot build thousands of real chips, so we **simulate** them. The method is called **Monte Carlo** (after the casino — it is all about random draws). Each virtual chip gets its own random mismatch, then we measure it, then we count how many pass.

Remember the three “PVT” villains? We make the rehearsal hard by testing the **full grid**:

Axis	Values	Count
Process corner	tt, ss, ff, sf, fs (first letter NMOS, second PMOS; t = typical, s = slow, f = fast)	5
Supply voltage	1.62 V, 1.80 V, 1.98 V (nominal ±10 %)	3
Temperature	−20 °C, 27 °C, 85 °C	3
Conditions	5 × 3 × 3	**45**
Trials per condition	750 random chips (the real core: 40, because it simulates much slower)	750
Total per design	45 × 750	**33,750**

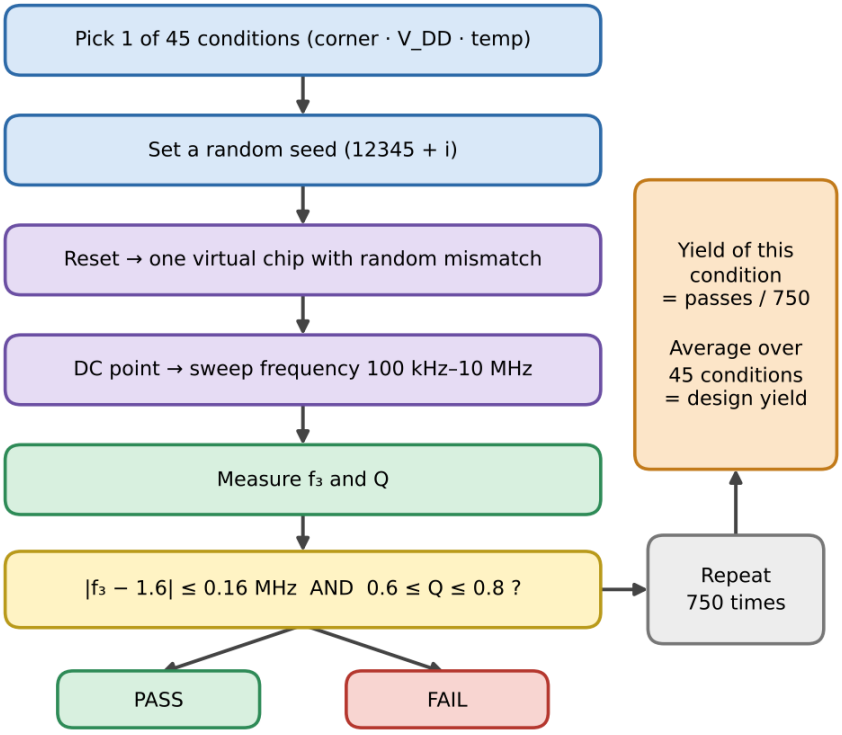


Figure 10 — The test loop for one condition. Repeat for all 45, then average the yields.

Measurement details, simply.

Quantity	How it is measured
f_3 (cutoff)	The frequency where the gain has dropped 3 dB below its DC value.
f_{90}	The frequency where the output is 90° late. Together with f_3: $\mathbf{Q = [(2 - (1 - u)^2) / u]^{-1/2}}$, $u = (f_3/f_{90})^2$.

Quantity	How it is measured
Pass rule	$\lvert f_3 - 1.6\ \text{MHz}\rvert \le 0.16\ \text{MHz}$ **and** $0.6 \le Q \le 0.8$. Ripple is reported, not part of the pass rule.
Yield	passes ÷ trials, per condition; the design yield is the mean over the 45 conditions.
Reproducibility	The seed for trial i is 12345 + i, so every run can be repeated exactly.
Uncertainty	Standard error $\sqrt{p(1-p)/N}$: ±3.6 points (one condition, N = 750), ±0.4 (45-condition mean); ±15 and ±2 for N = 40.

What we actually did (Chapter 6)
Wrote a Python driver that launches one ngspice 47 process per condition (45 in parallel). Each process seeds the random generator once, then repeats *reset → operating point → AC sweep (20 points/decade) → measure*. SKY130's "_mm" corners switch mismatch on for every transistor; there is no per-device switch, and a fresh draw needs a reset — which shaped the driver. Xyce 7.10 was used as an independent simulator.

CHAPTER 7

The Results

Who won, and by how much

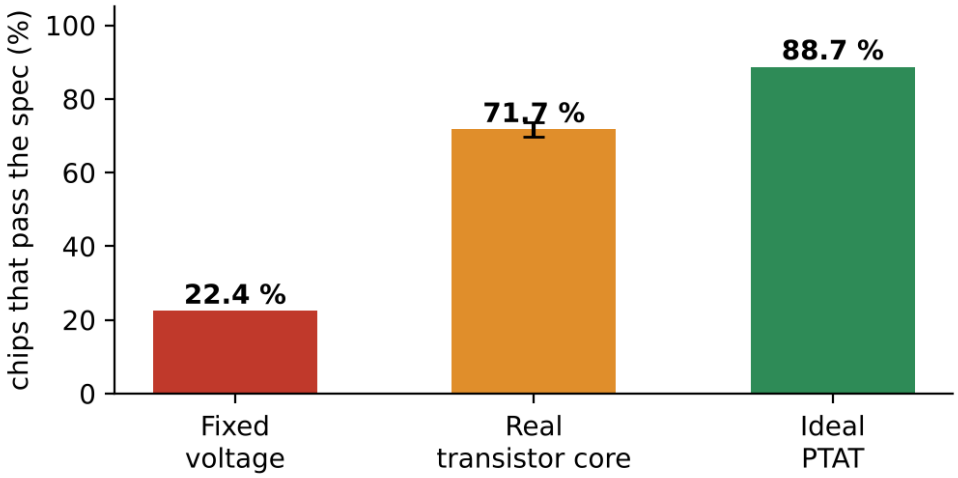


Figure 11 — The headline: yield (fraction of chips meeting the spec) for the three bias schemes.

Quantity	Fixed voltage	Ideal PTAT	Real core
Trials per condition	750	750	40
Mean yield	22.4 %	**88.7 %**	71.7 ± 2 %
Lowest-condition yield	0.0 %	72.3 %	17.5 %
Worst f_3 shift from 1.6 MHz	0.944 MHz	**0.084 MHz**	0.285 MHz
Mean $\sigma(f_3)$	5.32 %	6.07 %	6.75 %
Range of mean Q	0.598–0.806	0.697–0.737	0.690–0.752
Power	28.0–132.5 µW	72.0–113.0 µW	n/a
Yield at tt, 1.8 V, 27 °C	95.5 %	92.3 %	85.0 %

Reading the table like a story:

• **Fixed voltage** is brilliant exactly where it was tuned (95.5 % at tt, 1.8 V, 27 °C — slightly better than PTAT there) and collapses to 0.1 % once the temperature moves either way. That is why *single-point* simulation misleads. Its heat map is almost entirely dark except where corner and temperature accidentally cancel (ss at 85 °C, ff at −20 °C).

• **Ideal PTAT** keeps the mean cutoff within 5 % at every one of the 45 conditions and never falls below 72.3 % yield.

• **Fixed voltage's loss** is corner and temperature, not supply: its yield by supply is 22.8 / 22.3 / 22.2 % at 1.62 / 1.80 / 1.98 V — flat.

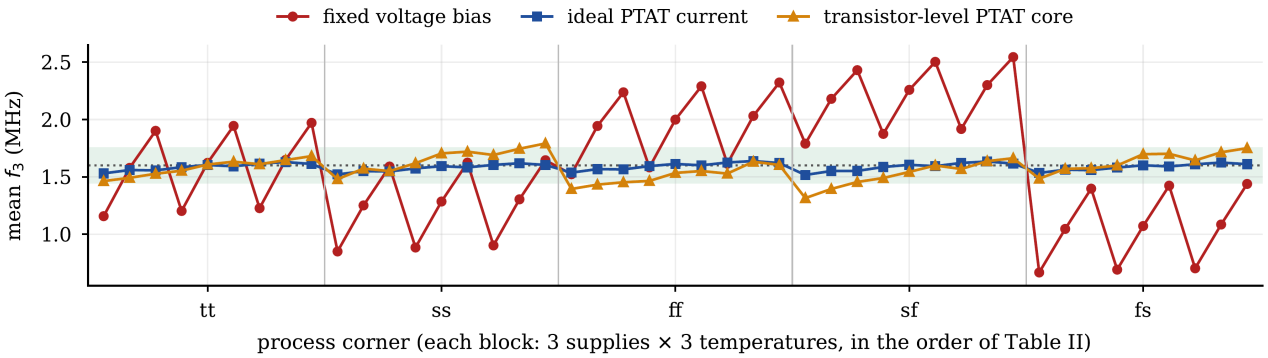


Figure 12 — Condition-mean cutoff across all 45 conditions. The red line (fixed voltage) swings wildly; the blue (ideal PTAT) hugs 1.6 MHz.

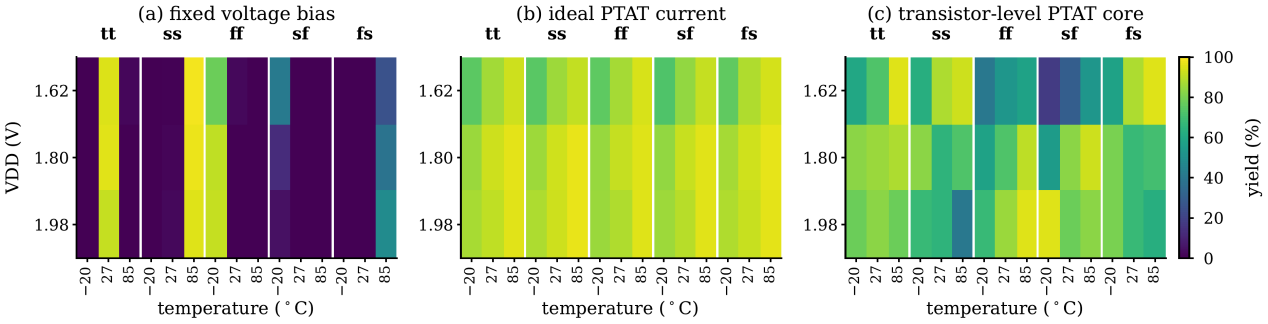


Figure 13 — Yield heat maps (yellow = good, dark = bad): (a) fixed voltage, (b) ideal PTAT, (c) real core.

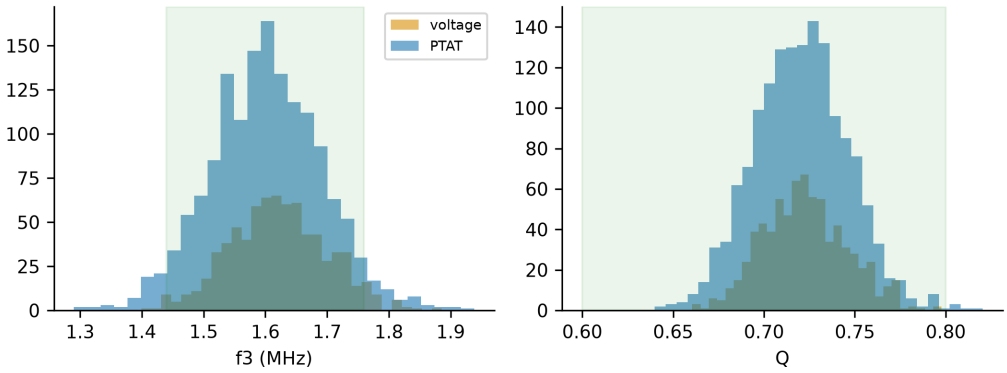


Figure 14 — Histograms at tt, 1.8 V, 27 °C: the random mismatch cloud for PTAT (blue) and fixed voltage (orange), against the spec window (green).

Mismatch alone, at the typical condition with PTAT bias, gives σ(f₃) = 5.64 % and σ(Q) = 0.026, a roughly Gaussian bell with 92.7 % inside the window. At the worst condition the bell moves to a 1.515 MHz mean and widens to 7.79 %, giving 73.0 %. The cloud mostly *translates* with the corner rather than changing shape.

What we actually did (Chapter 7)
Ran all 33,750 × 2 trials for the two 750-run designs plus 1,800 core runs, summarised every condition, and cross-checked every number in the paper against the result tables (this caught and fixed one transcription error in the by-supply yields).

CHAPTER 8

Why the Real Core Lost 17 Points

The gap between 88.7 % and 71.7 %

The ideal source ignores the supply. The real beta-multiplier does not: its current moves from 11.3 to 12.3 µA across the supply range, about **0.5 % current for every 1 % supply change**. That moves the mean cutoff from 1.49 through 1.61 to 1.65 MHz. Add the slightly-too-steep measured exponent (0.66 instead of 0.6) at low supply and low temperature, and the mean falls out of the window — exactly where the worst cell, **17.5 % (sf, 1.62 V, −20 °C)**, sits.

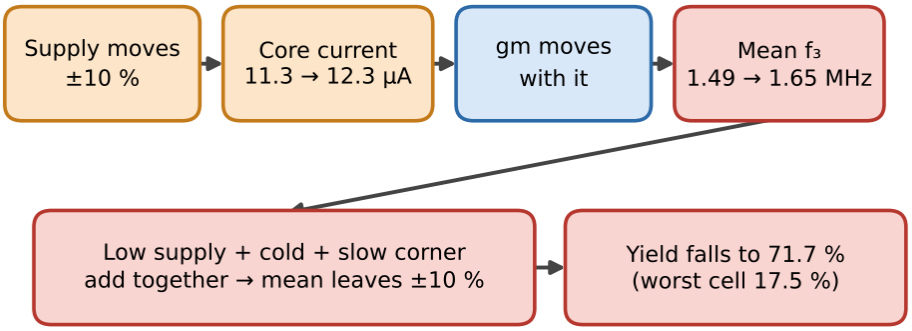


Figure 15 — How supply sensitivity turns into lost yield.

Group	Ideal / fixed	Real core
By supply (1.62 / 1.80 / 1.98 V)	Fixed: 22.8 / 22.3 / 22.2 % — flat	**64.0 / 76.2 / 74.8 %** — low-supply dip
By temperature (−20 / 27 / 85 °C)	—	65.5 / 72.0 / 77.5 % — varies much less

So the real core's problem is its **supply sensitivity, not temperature**. Lengthening its devices did not help. The next step is a supply-insensitive core — cascoded or regulated — rather than more area.

Does making the mirror bigger help?

We scaled the reference leg's area to see what Pelgrom buys:

Mirror size	σ(f₃)	Yield effect	Power
×1	5.91 %	baseline	90.9 µW
×4	4.81 %	+5 points at 27 °C, +6.6 at −20 °C	155.7 µW (+70 %)
×16	no further gain	none	≈ 4.6× baseline

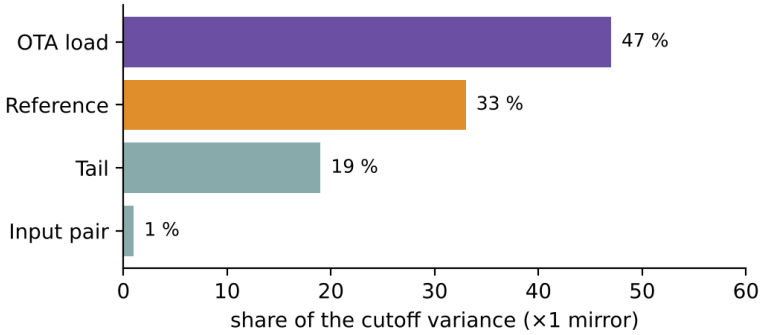


Figure 16 — Where the variance comes from at ×1. At ×4 the reference share falls to 11 % and the OTA load rises to 62 % — it becomes the limit, which is why ×16 is wasted.

CHAPTER 9

Did We Cheat Ourselves?

Five independent checks

A good scientist assumes they are wrong until shown otherwise. We ran five checks to separate "what the data say about the circuit" from "what they say about our tools".

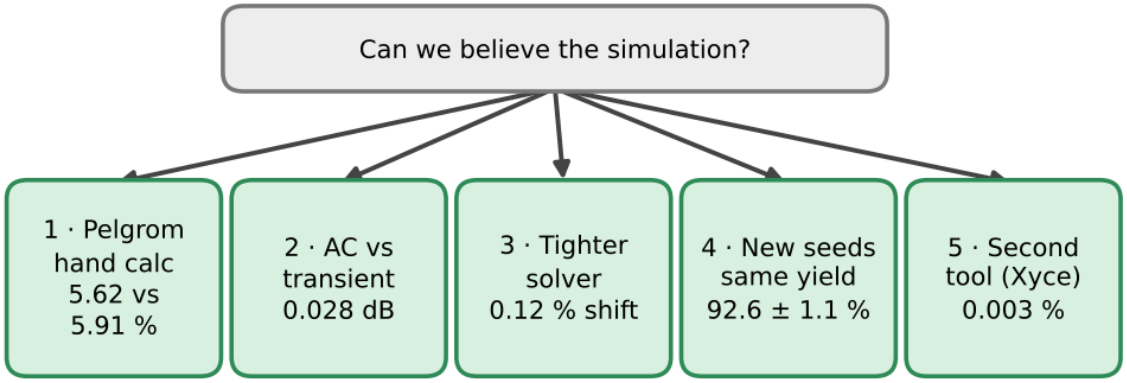


Figure 17 — The five lines of defence.

#	Check	What we found
1	Pelgrom hand calculation	σ(f₃) 5.62 % vs 5.91 % (MC); σ(Q) 3.61 % vs 3.68 %
2	AC analysis vs transient simulation	Agree to ≤ 0.028 dB and ≤ 0.14°
3	Tighter solver tolerance; coarse-grid effect	f₃ shifts 0.12 % (max 0.18 %); yield unchanged at 92.7 %
4	Different random seeds	92.3 ± 1.9 % (750 trials) vs 92.7 ± 1.3 % (1500); pooled 92.6 ± 1.1 %
5	Second simulator (Xyce)	Nominal agrees to 0.001 %; with the same injected offsets per-trial f₃ agrees to 0.003 %, correlation 1.00000

An honest catch
Xyce **ignores the PDK's mismatch model** — five seeds all returned the identical nominal answer. So we injected identical random gate-voltage offsets into 17 devices in both simulators. That proves the *simulators* agree. It does **not** prove the PDK's mismatch numbers are right — those have no published silicon validation. This is the most important limit on every yield number here.

Secondary measurements (so the filter is a real filter, not only a number): output noise 240 µV rms (1 kHz–10 MHz, ≈ 50 dB SNR for a 100 mV peak input); THD 0.33 ± 0.06 % at 200 kHz and 100 mV; 1 dB compression 65 mV at 1.6 MHz; step overshoot 4.76 ± 1.08 % (ideal Butterworth: 4.3 %); peaking 0.033 dB typical, 0.350 dB worst. A 5 % error in I_{ref} costs 5–12 yield points and a 5 % error in the 1.48 ratio costs 8–10, so the two legs must match to a few percent.

CHAPTER 10

What We Learned, and What Is Next

The moral of the story

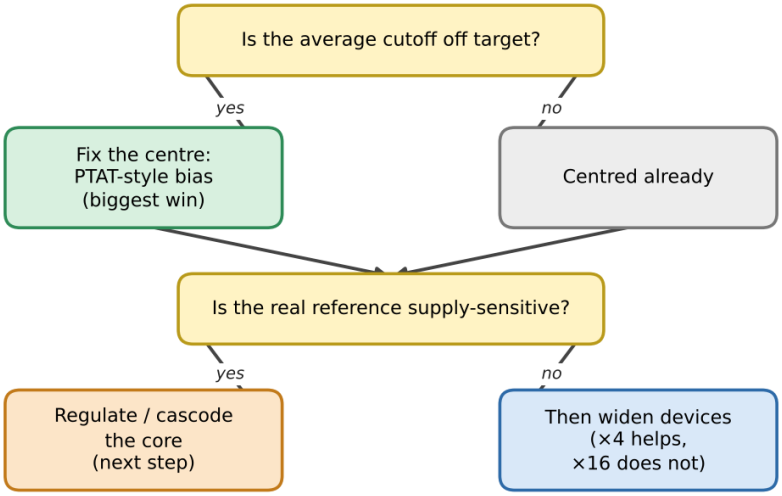


Figure 18 — The design guide, as a decision flow.

Four lessons

Lesson	Why
Centre first, narrow second	The systematic shift is about ten times the random spread. Bias compensation recovers most of the yield; matching area recovers only a few more points.
Match the exponent; budget the reference's supply sensitivity	An exponent near 0.6 suits this OTA (α = 1 overshoots, ≈ 79 % estimated). 0.5 % current per 1 % supply costs 17 yield points.
Size the mirror by yield per microwatt	×4 gives 5–7 points for 70 % more power; ×16 gives nothing.
Quote yields with intervals	750 trials: ±0.4 on a 45-condition mean, ±3.6 on one condition; 40 trials: ±2 and ±15.

The honest limits

• Pre-layout only: no parasitics, no package or supply noise.

• SKY130 mismatch and flicker coefficients are not validated against silicon; subthreshold modelling is documented as weak.

• The 88.7 % figure uses an ideal source — an upper bound. The real-core result rests on 40 trials per condition (per-condition values ±15 points; the mean ±2).

• No common-mode feedback and no output buffer, so the output's resting level is unregulated.

• All yields depend on the chosen window; a different spec moves all three numbers, though not their ordering.

The next step: a supply-regulated reference, then common-mode feedback, an output buffer and post-layout extraction on the way to a tape-out-ready design.

What was delivered

Deliverable	What it is
Project report	56 pages, 30 figures, in the supplied college format
Final review deck	37 slides: objective, literature review, circuit, results, live demo, output explanation
IEEE conference paper	6 pages (INDICON limit: 4–6 including references) with PDF, LaTeX source package and a Word version
Open artifacts	Netlists, seeds, drivers and result tables (repository link still to be added to the paper)

Two-minute version to tell someone
“We built a small filter in a free chip process and asked how many copies would actually work after the factory, the supply and the temperature all vary, and after random transistor mismatch. With a plain voltage bias only 22 % work. If the bias current rises with temperature — PTAT — about 89 % work with an ideal source and about 72 % with a real circuit; the rest is lost because the real reference is sensitive to the supply. The random mismatch spread, about 5–7 %, cannot be removed by biasing — only shrunk with area. We checked it five different ways.”

GLOSSARY

Every term, one line each

Term	Meaning
Biquad	Second-order filter (two storage elements) — a steeper roll-off than a single RC.
Butterworth	The flattest passband, no bump; needs Q = 0.707.
Corner (tt, ss, ff, sf, fs)	Extreme factory cases. First letter NMOS, second PMOS; t typical, s slow, f fast.
Cutoff frequency (f_3)	Where the output has fallen 3 dB (≈ 70 % amplitude). Target 1.6 MHz ±10 %.
dB	Log gain scale; −3 dB ≈ 0.707 in amplitude.
Differential pair	Two matched transistors that amplify the difference between two inputs — the OTA input stage.
gm (transconductance)	Voltage in → current out; here ≈ 100 µS. The “strength” of an amplifier.
Gm-C filter	A filter built from transconductors and capacitors, with no resistors.
Mismatch	Small random differences between nominally identical devices.
Mobility	How easily carriers move; falls when hot.
Monte Carlo	Repeating a simulation with random draws to see the spread of results.
Operating point	The DC state of the circuit before analysing it.
OTA	Operational Transconductance Amplifier — the “tap”.
PDK / SKY130	Process Design Kit: the factory’s transistor models. SKY130 is SkyWater’s open 130 nm PDK.
Pelgrom law	$\sigma(\Delta V_T) = A / \sqrt{(W \cdot L)}$: mismatch falls with the square root of area.
Phase	Delay of the output wave relative to the input, in degrees.
PTAT	Proportional To Absolute Temperature — a current that rises as it gets hotter.
PVT	Process, Voltage, Temperature — the systematic variations.
Q (quality factor)	How peaky the filter is near cutoff. Spec 0.6–0.8.
Beta-multiplier	A self-biasing current reference whose current depends on a resistor and mobility, to first order not on the supply.
Seed	Starting value of a random generator; same seed → same random draws.
σ (sigma)	Standard deviation — the typical spread around the average.
Sensitivity	How much an output changes per unit change of one input.
Tail current	The current source under the differential pair — the knob that sets gm.
Threshold voltage V_T	The gate voltage at which a transistor starts conducting.
THD	Total harmonic distortion — how much the output wave is distorted.
Xyce / ngspice	The two circuit simulators used.
Yield	Fraction of trials meeting the spec (f_3 in window and Q in window).